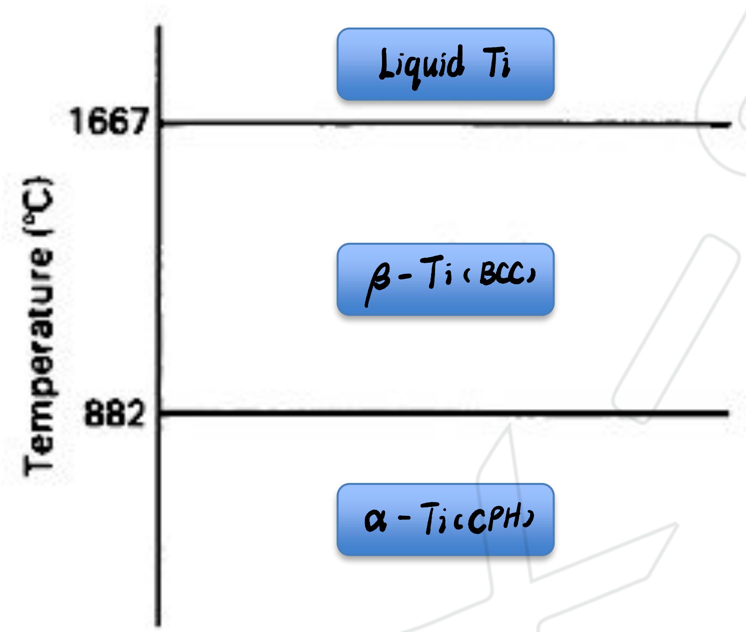
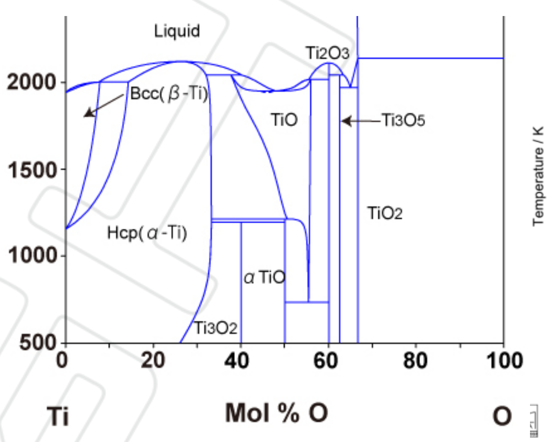
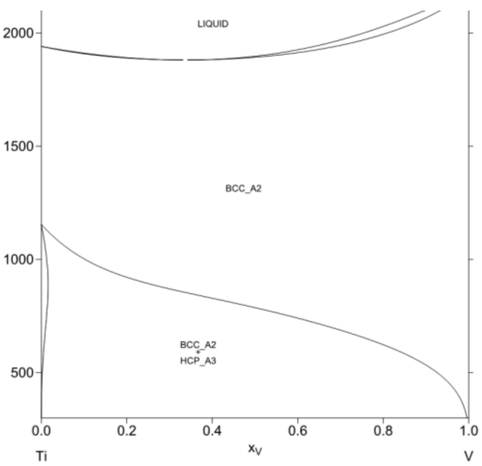
2. Alloys: Ti

1. **Titanium (Ti)**
   Ti is a lightweight metal with many advantages for engineering application. Ti is a ___ploymorphic___ metal. Label the different phases of Ti at different temperatures and indicate their lattice packing in the figure below.



By adding different alloy elements, the phase transformation temperature can be changed. According to the phase-diagram (below left) of Ti-O, O is an ___α stabiliser___; while the phase-diagram of Ti-V (below right) indicates that V is a ___β stabiliser___. Apart from O, ___C___ and ___N___ can also act as α stabiliser. These small atoms form interstitial solid solutions, ___interstitial soild solution___ strengthening is much more effective than substitutional solid solution strengthening.





2. Ti alloys

Fill the table below accordingly.

Ti alloys	α alloy	Near α alloy	α-β alloy	Near β alloy	β alloy
Example alloys	C.P. Ti-5Al-2.5Sn	Ti-6Al-2Sn-4Zr-2Mo	Ti 64	Ti-10V-2Fe-3Al	Ti-13V-11Cr-3Al
Equilibrium constitution at T_{room}	Single-phase alloy α-Ti	Engineering interesting area is α+ β	α + β	α+ β	β with few% α
Properties	Not engineering relevant	Best creep resistance among all Ti alloys	A good balance of fatigue and creep resistance at elevated T	Very good strength at low temperature	Highest strength Ti alloy
Strengthening mechanisms *	SS, GB, WH	SS; GB	SS; GB; Ppts	SS ; GB ; ppts	SS; GB; ppts
Application	Not engineering relevant	Aeroengine compressor blade	Fan blade of aeroengine	Landing gear	//////

- SS: solid solution strengthening;
- GB: grain boundary strengthening;
- WH: work hardening;
- Ppts: Precipitation strengthening.

Learn more detailed explanation in the lecture slides.

3. Summary of aerospace Ti alloys